10_Symmetric & Asymmetric Matrix

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Symmetric :

A symmetric matrix is a **square matrix** that is **equal to its transpose.**

$$A^T = A$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \longrightarrow A^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Properties:

- Always square $(n \times n)$.
- Main diagonal remains unchanged.
- Eigenvalues are always real.
- Used in PCA, covariance matrices, and physics applications.

Asymmetric :

An asymmetric matrix is a **square matrix** that is **not equal to its transpose**.

$$A^T \neq A$$

$$B = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 3 & 6 \\ 7 & 8 & 9 \end{bmatrix} \longrightarrow B^T = \begin{bmatrix} 1 & 2 & 7 \\ 4 & 3 & 8 \\ 5 & 6 & 9 \end{bmatrix}$$

Properties:

- Not necessarily square ($m \times n$ or $n \times n$).
- Rows and columns are different.
- Can have complex or real eigenvalues.
- Used in directed graphs, adjacency matrices, and transformations.

Importance of symmetric Matrix

- **Principal Component Analysis (PCA) & Machine Learning:**
  - Covariance matrices (used in PCA) are symmetric, ensuring real eigenvalues that help in dimensionality reduction.
  - Kernel matrices in support vector machines (SVM) are symmetric, aiding in efficient classification.
- **Symmetric matrices require storing only half the elements, reducing memory usage.**
- **Graph Theory & Networks**
- **Applications in Physics & Engineering**

Comment below whether a matrix is symmetric or asymmetric.

$$A = \begin{bmatrix} 4 & 2 & -1 \\ 2 & 3 & 5 \\ -1 & 5 & 6 \end{bmatrix} \qquad B = \begin{bmatrix} 7 & 1 & 3 \\ 1 & 5 & 2 \\ 3 & 2 & 8 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

- Matrix A: Symmetric ✅
- Matrix B: Symmetric ✅
- Matrix C: Asymmetric ❌